GCSE PHYSICS

Higher Tier – Paper 1 style

Topic 4.1: Energy

Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 165 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **158**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 158

0 1 This question is about energy stores and energy transfers.

0 1 . 1 Choose the energy store that **increases** in each situation. Write one word or phrase from the box in each row. **[4 marks]**

kinetic gravitational potential elastic potential thermal

Situation	Energy store that increases
A hot cup of tea cools down	
A ball is thrown upwards	
A spring is squashed	
A car speeds up	

0 1 . 2 Name the **four** ways energy can be transferred between stores. **[4 marks]**

0 1 . 3 A kettle transfers energy from the mains. State the energy transfer when the kettle heats water. **[2 marks]**

0 2 A car has a mass of 1200 kg.

0 2 . 1 Write down the equation that links kinetic energy, mass and speed. **[1 mark]**

0 2 . 2 The car travels at 15 m/s. Calculate the kinetic energy of the car. **[3 marks]**

Answer = _____ J

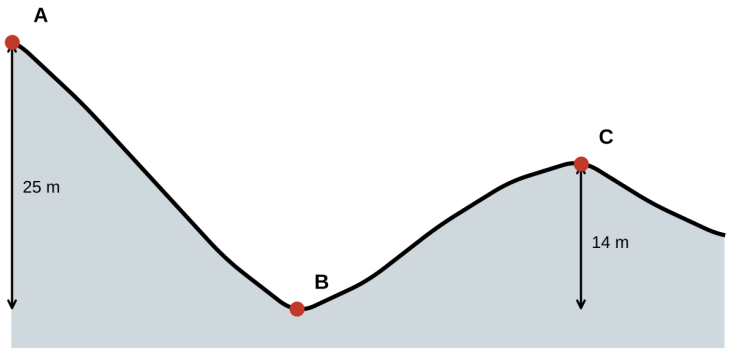
0 2 . 3 The car speed doubles to 30 m/s. Calculate the kinetic energy and state how it compares with the answer above. **[3 marks]**

Answer = _____ J

0 2 . 4 A trolley has a kinetic energy of 500 J and a mass of 4.0 kg. Calculate the speed of the trolley. **[4 marks]**

Answer = _____ m/s

0 3 A roller coaster car and its passengers have a total mass of 450 kg. The diagram shows part of the track. The car starts from rest at **A**, 25 m above the lowest point **B**. gravitational field strength = 9.8 N/kg.



0 3 . 1 Write down the equation that links gravitational potential energy, mass, gravitational field strength and height. **[1 mark]**

0 3 . 2 Calculate the gravitational potential energy gained by the car when it is lifted from **B** to **A**. **[3 marks]**

Answer = _____ J

0 3 . 3 Assume no energy is wasted as the car rolls down from **A** to **B**. State the maximum kinetic energy of the car at **B**. **[1 mark]**

0 3 . 4 Calculate the speed of the car at **B**. **[4 marks]**

Answer = _____ m/s

0 3 . 5 Calculate the gravitational potential energy of the car at **C**, 14 m above **B**. **[2 marks]**

Answer = _____ J

0 3 . 6 In practice the car is slower at **B** than the value calculated. Explain why. **[2 marks]**

0 4 A student compresses a spring and uses it to launch a toy. The spring constant of the spring is 40 N/m.

0 4 . 1 Write down the equation that links elastic potential energy, extension and spring constant. **[1 mark]**

0 4 . 2 The spring is compressed by 0.15 m. Calculate the elastic potential energy stored in the spring. **[3 marks]**

Answer = _____ J

0 4 . 3 All the energy is transferred to the kinetic energy store of a 0.050 kg toy. Calculate the maximum speed of the toy. **[4 marks]**

Answer = _____ m/s

0 4 . 4 State why the speed of the toy is actually less than your answer. **[1 mark]**

0 5 The specific heat capacity of water is 4200 J/kg °C.

0 5 . 1 Write down the equation that links change in thermal energy, mass, specific heat capacity and temperature change. **[1 mark]**

0 5 . 2 Define **specific heat capacity**. **[1 mark]**

0 5 . 3 Calculate the energy needed to heat 0.50 kg of water from 20 °C to 60 °C. **[3 marks]**

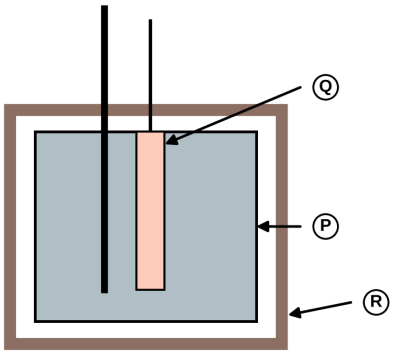
Answer = _____ J

0 5 . 4 An electric kettle has a power of 2.3 kW. Calculate the time taken for the kettle to supply the energy in part 3, assuming no energy is wasted. **[4 marks]**

Answer = _____ s

0 5 . 5 Explain why the real time taken is longer than the time you calculated. **[2 marks]**

0 6 A student investigates the specific heat capacity of an aluminium block. The diagram shows the apparatus. The block has a mass of 1.0 kg and the heater has a power of 50 W.

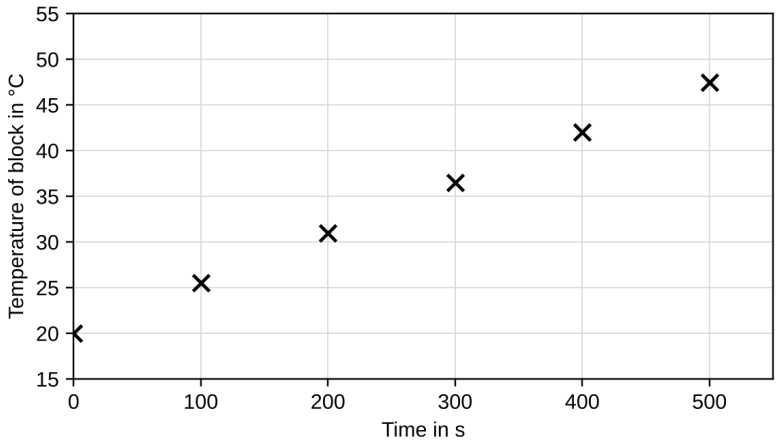


0 6 . 1 Name the pieces of apparatus labelled **P**, **Q** and **R**. **[3 marks]**

0 6 . 2 Explain why the block is wrapped in insulation (R). **[2 marks]**

0 6 . 3 State **one** way the student can make sure the block warms evenly and the thermometer reading is accurate. **[1 mark]**

0 7 The student's results are plotted on the graph below. The heater power is 50 W and the block has mass 1.0 kg.



0 7 . 1 Draw a line of best fit on the graph. **[1 mark]**

0 7 . 2 Calculate the energy transferred to the block in the first 500 s. **[3 marks]**

Answer = _____ J

0 7 . 3 Use the graph to find the temperature rise in 500 s, and calculate the specific heat capacity of the block. **[4 marks]**

Answer = _____ J/kg °C

0 7 . 4 The accepted value for aluminium is 900 J/kg °C. Suggest why the student's value is close to this. **[1 mark]**

0 8 This question is about power.

0 8 . 1 Write down the **two** equations that can be used to calculate power. **[2 marks]**

0 8 . 2 A hairdryer transfers 180 000 J of energy in 120 s. Calculate its power. **[3 marks]**

Answer = _____ W

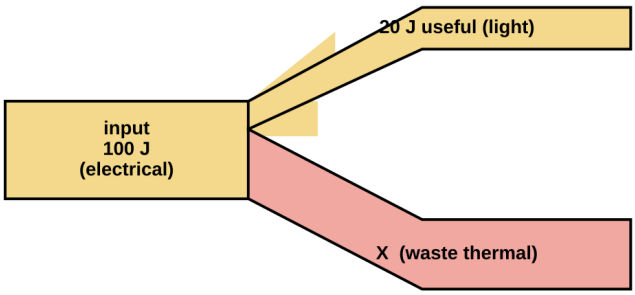
0 8 . 3 A 60 W lamp is on for 2 hours. Calculate the energy transferred in joules. **[4 marks]**

Answer = _____ J

0 8 . 4 A student of mass 55 kg climbs a flight of stairs 3.0 m high in 5.0 s. Calculate the useful power output of the student. gravitational field strength = 9.8 N/kg. **[4 marks]**

Answer = _____ W

0 9 The Sankey diagram shows the energy transfers in a filament lamp.



0 9 . 1 Calculate the amount of wasted energy, **X**. **[1 mark]**

0 9 . 2 State what happens to the wasted energy. **[2 marks]**

0 9 . 3 Write down the equation for efficiency. **[1 mark]**

0 9 . 4 Calculate the efficiency of the filament lamp. Give your answer as a percentage. **[3 marks]**

Answer = _____ %

0 9 . 5 An LED lamp is 85% efficient. For 40 J of input energy, calculate the useful energy output. **[3 marks]**

Answer = _____ J

0 9 . 6 A student says: 'This device is 120% efficient.' Explain why this is impossible. **[2 marks]**

1 0 This question is about the conservation of energy.

1 0 . 1 State the principle of conservation of energy. **[2 marks]**

1 0 . 2 What is meant by a **closed system**? **[1 mark]**

1 0 . 3 A pendulum is released from rest and swings back and forth. Each swing is slightly lower than the last. Explain what happens to the energy of the pendulum. **[4 marks]**

1 0 . 4 A ball is dropped from a height of 2.0 m. Explain the energy transfers as it falls and bounces. **[3 marks]**

1 1 Energy transfers can be wasted in machines and in buildings.

1 1 . 1 Give **two** ways of reducing unwanted energy transfers in a machine. **[2 marks]**

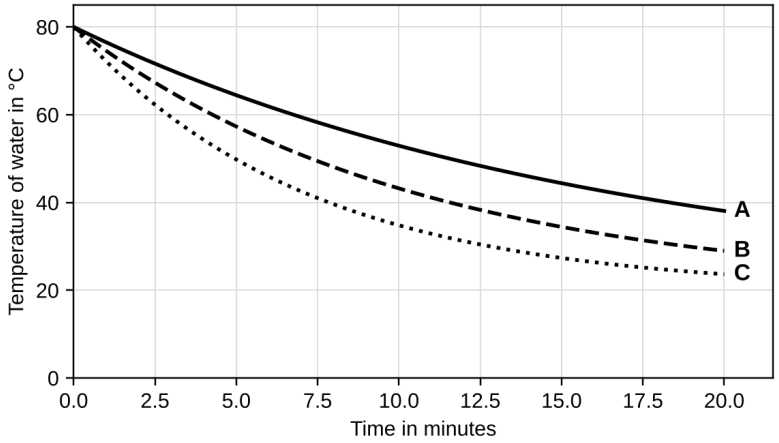
1 1 . 2 Explain how lubricating a bicycle chain improves its efficiency. **[2 marks]**

1 1 . 3 State how the thermal conductivity of the walls of a house affects the rate of energy transfer by conduction. **[1 mark]**

1 1 . 4 A house owner wants to reduce heating bills. Give **three** ways she could reduce energy loss. **[3 marks]**

1 1 . 5 Explain why a thicker wall reduces the rate of energy transfer. **[2 marks]**

1 2 A student investigates insulation. She fills three identical beakers with 200 cm^3 of hot water. Beaker **A** is wrapped in thick foam, beaker **B** in thin foam, and beaker **C** has no insulation. She records the temperature every minute. The graph shows her results.



1 2 . 1 Which beaker, **A**, **B** or **C**, was wrapped in the thick foam? Give a reason for your answer. **[2 marks]**

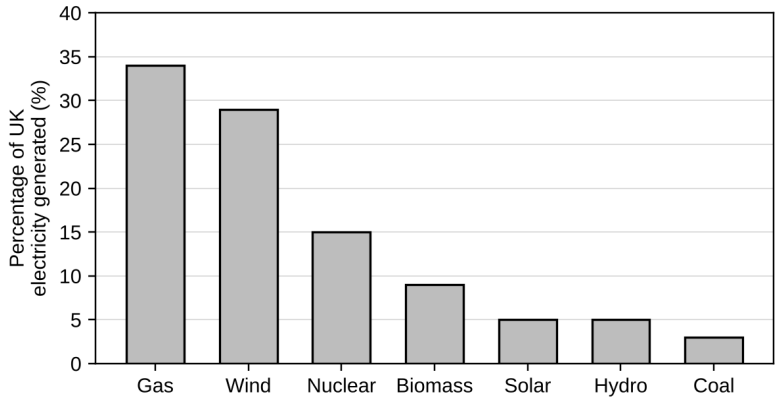
1 2 . 2 Give **two** variables the student should control. **[2 marks]**

1 2 . 3 Use the graph to find the temperature of beaker **C** after 10 minutes. **[1 mark]**

1 2 . 4 Describe how the rate of cooling of beaker **A** changes with time and explain the change. **[3 marks]**

1 2 . 5 The student repeats the experiment with a shiny foil covering. Predict how the cooling curve changes and explain your prediction. **[2 marks]**

1 3 The bar chart shows how UK electricity was generated in one year.



1 3 . 1 Which energy resource generated the most electricity? **[1 mark]**

1 3 . 2 Calculate the total percentage of electricity generated from renewable resources. **[3 marks]**

Answer = _____ %

1 3 . 3 Give **two** disadvantages of using wind turbines. **[2 marks]**

1 3 . 4 Explain one advantage of using nuclear power rather than gas. **[2 marks]**

1 3 . 5 State what is meant by a **non-renewable** energy resource. **[1 mark]**

1 3 . 6 A power station has an input of 500 MW and is 40% efficient. Calculate the useful power output. **[2 marks]**

Answer = _____ MW

1 4 Read the question and answer in full sentences.

1 4 . 1 *Two students want to find out which material is the best thermal insulator for a house wall. Describe a method they could use to compare materials A, B and C. Include how they would make it a fair test and how they would use the results. **[6 marks]**

1 4 . 2 *Evaluate the use of solar cells, wind turbines and a gas power station to supply electricity for a village in the UK. **[6 marks]**

1 5 A student uses a motor to lift a 2.0 kg mass through a height of 1.5 m in 3.0 s. The motor takes 40 J of electrical energy. gravitational field strength = 9.8 N/kg.

1 5 . 1 Calculate the gravitational potential energy gained by the mass. **[3 marks]**

Answer = _____ J

1 5 . 2 Calculate the efficiency of the motor. **[3 marks]**

Answer = _____ %

1 5 . 3 Calculate the useful power output of the motor. **[3 marks]**

Answer = _____ W

1 5 . 4 Calculate the energy wasted by the motor and state where the energy goes. **[2 marks]**

_____

_____

_____

END OF QUESTIONS

MARK SCHEME – Topic 4.1: Energy

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	thermal ; gravitational potential ; elastic potential ; kinetic ;	One mark each. Tea: accept 'thermal store of the surroundings'.	4	AO1 4.1.1.1
01.2	mechanically (by a force doing work) ; electrically (work done by moving charges) ; by heating ; by radiation (light / sound waves) ;	Allow 'heating by particles / conduction, convection' for heating. Allow 'waves' for radiation.	4	AO1 4.1.1.1
01.3	electrically ; to the thermal energy store of the water ;		2	AO1 4.1.1.1
02.1	kinetic energy = 0.5 × mass × (speed)^2		1	AO1 4.1.1.2
02.2	E_k = 0.5 × 1200 × 15^2 ; = 0.5 × 1200 × 225 ; 135 000 (J) ;	Correct answer with or without working scores 3 marks. Common error: not squaring the speed (gives 9000 J, 2 marks max).	3	AO2 4.1.1.2
02.3	E_k = 0.5 × 1200 × 30^2 = 540 000 J ; (four times) bigger ; doubling the speed makes the kinetic energy four times larger ;	Allow ecf from part 2.	3	AO3 4.1.1.2
02.4	500 = 0.5 × 4.0 × v^2 ; v^2 = 500 ÷ 2.0 = 250 ; v = √250 ; 15.8 (m/s) ;	Allow 16 (m/s). 15.81... scores all 4 marks.	4	AO2 4.1.1.2
03.1	gravitational potential energy = mass × gravitational field strength × height		1	AO1 4.1.1.2
03.2	E_p = 450 × 9.8 × 25 ; = 4410 × 25 ; 110 250 (J) ;	Allow 110 000 J or 1.1 × 10^5 J.	3	AO2 4.1.1.2
03.3	110 250 (J) (ecf from part 2)		1	AO2 4.1.1.3
03.4	110 250 = 0.5 × 450 × v^2 ; v^2 = 490 ; v = √490 ; 22.1 (m/s) ;	Allow ecf from part 3. Allow 22 m/s.	4	AO2 4.1.1.2
03.5	450 × 9.8 × 14 ; 61 740 (J) ;		2	AO2 4.1.1.2
03.6	some energy is wasted / dissipated ; by friction between the wheels and track (and air resistance), transferring energy to thermal stores of the surroundings ;		2	AO2 4.1.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	elastic potential energy = 0.5 × spring constant × (extension)^2		1	AO1 4.1.1.2
04.2	$E_e = 0.5 \times 40 \times 0.15^2$; $= 20 \times 0.0225$; 0.45 (J) ;	Common error: using 15 (cm) instead of 0.15 m (gives 4500 J, 2 marks max).	3	AO2 4.1.1.2
04.3	$0.45 = 0.5 \times 0.050 \times v^2$; $v^2 = 18$; $v = \sqrt{18}$; 4.2 (m/s) ;	Allow 4.24 m/s. Allow ecf from part 2.	4	AO2 4.1.1.2
04.4	some energy is wasted / transferred to thermal stores by friction / air resistance		1	AO2 4.1.1.3
05.1	change in thermal energy = mass × specific heat capacity × temperature change		1	AO1 4.1.1.3
05.2	the energy needed to raise the temperature of 1 kg of a substance by 1 °C		1	AO1 4.1.1.3
05.3	$\Delta E = 0.50 \times 4200 \times 40$; $= 84\,000$ (J) ; (unit) J / 84 kJ ;	Third mark for the correct unit with the correct number only if consistent (J or kJ).	3	AO2 4.1.1.3
05.4	power = 2300 W ; $t = E \div P$; $= 84\,000 \div 2300$; 36.5 (s) ;	Allow 37 s. Allow ecf from part 3. Common error: not converting kW to W (0.0365 s, 3 marks max).	4	AO2 4.1.1.4
05.5	some energy is transferred to the surroundings / kettle itself ; so more energy must be supplied / not all electrical energy goes to the water ;		2	AO2 4.1.1.4
06.1	P: (aluminium) block ; Q: (electric / immersion) heater ; R: insulation / lagging ;	One mark each.	3	AO1 4.1.1.3 RP
06.2	to reduce energy transferred to the surroundings ; so (almost) all the energy supplied goes into the block / the result is more accurate ;		2	AO2 4.1.1.3 RP
06.3	put a little oil / water in the thermometer hole (to improve thermal contact) ; wait for the reading to stop rising before recording / read thermometer at eye level ;	Any one.	1	AO3 4.1.1.3 RP
07.1	straight line through the points (ruler, not forced through the origin) ;		1	AO3 WS 3.3
07.2	$E = P \times t$; $= 50 \times 500$; 25 000 (J) ;		3	AO2 4.1.1.4
07.3	temperature rise = 47.5 − 20 = 27.5 (°C) ; $c = \Delta E \div (m \times \Delta\theta)$; $= 25\,000 \div (1.0 \times 27.5)$; 909 (J/kg °C) ;	Allow 900 J/kg °C. Allow values 880–940.	4	AO3 4.1.1.3 RP
07.4	the block was well insulated so little energy was lost / heater transferred most energy to the block		1	AO3 4.1.1.3 RP

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.1	power = energy transferred ÷ time ; power = work done ÷ time ;	Allow P = E / t and P = W / t.	2	AO1 4.1.1.4
08.2	P = 180 000 ÷ 120 ; 1500 ; W ;		3	AO2 4.1.1.4
08.3	time = 2 × 3600 = 7200 s ; E = P × t ; = 60 × 7200 ; 432 000 (J) ;	Allow 432 kJ. Common error: using 2 hours in the equation (120 J).	4	AO2 4.1.1.4
08.4	E = 55 × 9.8 × 3.0 = 1617 J ; P = 1617 ÷ 5.0 ; 323 (W) ;	Allow 320 W. Allow 1 mark for each of the three steps, last mark for the answer in W.	4	AO3 4.1.1.4
09.1	80 (J)		1	AO2 4.1.2.2
09.2	transferred to the thermal energy store of the surroundings ; it is dissipated / spreads out and is not usefully stored ;		2	AO1 4.1.2.2
09.3	efficiency = useful output energy transfer ÷ total input energy transfer		1	AO1 4.1.2.2
09.4	20 ÷ 100 ; 0.20 ; 20 (%) ;		3	AO2 4.1.2.2
09.5	0.85 × 40 ; 34 ; J ;	Allow 34 for 2 marks without a unit.	3	AO2 4.1.2.2
09.6	efficiency cannot be above 100% ; energy cannot be created / useful output cannot be greater than the input (conservation of energy) ;		2	AO2 4.1.2.1
10.1	energy cannot be created or destroyed ; only transferred from one store to another / total energy before = after ;		2	AO1 4.1.2.1
10.2	a system where no energy (or matter) enters or leaves / no net change in total energy		1	AO1 4.1.2.1
10.3	energy is transferred from gravitational potential to kinetic and back ; some energy is transferred to thermal stores by friction / air resistance ; this energy is dissipated to the surroundings ; total energy is conserved but the pendulum has less useful energy so swings lower ;		4	AO3 4.1.2.1
10.4	gravitational potential → kinetic as it falls ; kinetic → elastic potential during the bounce, then back to kinetic ; some energy is wasted as thermal / sound so it does not bounce back to the same height ;		3	AO2 4.1.2.1
11.1	lubrication (oil on moving parts) ; thermal insulation ; streamlining to reduce air resistance ;	Any two.	2	AO1 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
11.2	oil reduces friction between moving parts ; so less energy is transferred to thermal energy / more is transferred usefully ;		2	AO2 4.1.2.2
11.3	the higher the thermal conductivity, the faster the rate of energy transfer		1	AO1 4.1.2.2
11.4	cavity wall insulation ; loft insulation ; double glazing ; draught-proofing ;	Any three.	3	AO2 4.1.2.2
11.5	thicker wall means the energy has to travel further ; so the temperature gradient is smaller / rate of transfer is lower ;		2	AO2 4.1.2.2
12.1	A ; its temperature fell most slowly / least energy lost ;		2	AO3 4.1.2.2
12.2	starting temperature (of the water) ; volume / mass of water ; same beaker material / size ; room temperature / lid ;	Any two.	2	AO3 WS 2.2
12.3	36–40 (°C)		1	AO3 WS 3.1
12.4	rate of cooling decreases / graph gets less steep ; temperature difference between water and surroundings decreases ; so less energy is transferred per second ;		3	AO3 4.1.2.2
12.5	cools more slowly ; shiny surfaces are poor emitters of infrared radiation ;		2	AO3 4.1.2.2
13.1	gas		1	AO3 4.1.3
13.2	renewables are wind, biomass, solar, hydro ; 29 + 9 + 5 + 5 ; 48 (%) ;	Allow 1 mark for correctly identifying three renewable sources.	3	AO2 4.1.3
13.3	unreliable / depends on weather ; visual / noise pollution ; high set-up cost / cannot be turned on on demand ; harm to birds ;	Any two.	2	AO1 4.1.3
13.4	no carbon dioxide produced in generation ; so does not contribute to global warming / produces a lot of energy from small amounts of fuel ;		2	AO2 4.1.3
13.5	a resource that will run out / cannot be replaced as quickly as it is used		1	AO1 4.1.3
13.6	0.40 × 500 ; 200 (MW) ;		2	AO2 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
14.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6 marks): a clear, logical method with the main variables controlled and a clear conclusion linked to results. • Level 2 (3–4 marks): a method with some control of variables and some use of results. • Level 1 (1–2 marks): simple, incomplete statements. • Indicative content: wrap identical containers / beakers in each material; fill with the same volume of hot water at the same start temperature; record temperature at regular intervals with a thermometer / data logger; same thickness of each material; same room, same lid; measure temperature drop after a fixed time or plot cooling curves; the best insulator gives the smallest temperature drop / slowest cooling; repeat and calculate a mean.		6	AO3 4.1.2.2 / RP

Question	Answers	Extra information	Mark	AO / Spec. Ref.
14.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): a balanced evaluation, at least one advantage and one disadvantage for each source, and a justified judgement. • Level 2 (3–4): advantages / disadvantages for at least two sources, partial judgement. • Level 1 (1–2): simple statements about one or two sources. • Indicative content: solar – no emissions, free energy, no fuel; but only in daylight, high initial cost, weather dependent; wind – no emissions, low running costs; unreliable, noise, visual impact, depends on wind speed; gas – reliable, can respond to demand quickly; releases CO_2, non-renewable, fuel costs; combination or storage may be best; judgement such as 'wind and solar with a gas back-up'.		6	AO3 4.1.3
15.1	2.0 × 9.8 × 1.5 ; 29.4 ; J ;		3	AO2 4.1.1.2
15.2	29.4 ÷ 40 ; 0.735 ; 73.5 (%) ;	Allow ecf from part 1.	3	AO2 4.1.2.2
15.3	P = 29.4 ÷ 3.0 ; 9.8 ; W ;		3	AO2 4.1.1.4
15.4	40 − 29.4 = 10.6 J ; thermal store of the motor / surroundings (sound) ;		2	AO2 4.1.2.2

Where marks are lost in Topic 4.1

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Always write the equation first** (it often earns a mark), then substitute, then give the answer with the unit. Show every step.
- ☐ **Kinetic energy:** square the **speed** only, not 0.5 × mass × speed. A doubled speed gives four times the kinetic energy.
- ☐ **Units:** convert kW to W, minutes and hours to seconds, cm to m, g to kg **before** substituting.
- ☐ **Energy 'wasted'** is not destroyed: it is dissipated into the thermal store of the surroundings. Say this, never 'lost'.
- ☐ **Specific heat capacity:** $\Delta E = m\,c\,\Delta\theta$. For a heater, $\Delta E = P \times t$. Mass must be in kg.
- ☐ **Efficiency** has no units as a decimal; ×100 for a percentage. It can never be more than 100%.
- ☐ **Required practical:** insulate to reduce losses, use oil in thermometer hole, wait for the reading to stabilise, plot a straight line of best fit and use gradient.
- ☐ **Evaluate 6-mark answers:** give advantages and disadvantages for **each** option, then a justified judgement.